Policy

High Risk Medication

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Staff

Contents

Scope 1
Introduction 3
Risk Statement 3
Definitions 3
Policy Principles 4
Accountabilities and Responsibilities 4
Appendix 1. High Risk Medication Register 7
1. High Risk Medicines Management Strategies 7
2. Antimicrobials 7
   2.1 Aminoglycosides (Gentamicin, Tobramycin and Amikacin) 7
   2.2 Amphotericin 7
   2.3 Vancomycin 8
3. Potassium and Other Electrolytes 8
   3.1 Potassium 8
   3.1 Calcium 9
   3.2 Magnesium 9
   3.3 Phosphate 10
   3.4 Sodium Chloride – Hypertonic Saline 10
4. Psychotropic Agents 10
   4.1 Clozapine 11
   4.2 Lithium 11
   4.3 Zuclopenthixol Acetate 12
5. Insulin 12
   5.1 Insulin by Subcutaneous Injection 13
   5.2 Storage of Insulin 13
   5.3 Administering Insulin by Intravenous Infusion 14
6. Narcotics/Opioids and Sedative Agents 14

6.1 Prescribing and Administering Opioids and Sedative Agents 15
6.2 Patient Controlled Intravenous Analgesia 16
6.3 Narcotic Transdermal Patch Delivery Systems 16
6.4 Restricted Schedule 4 (S4R) Medicines 17
7. Neuromuscular Blocking Agents 17
7.1 Storage Requirements for NMBAs 17
7.2 Prescribing Requirements for NMBAs 18
7.3 Preparation and Administration of NMBAs 18
8. Chemotherapeutic Agents 18
8.1 Vinca Alkaloids 19
8.2 Intrathecal Chemotherapy 20
8.3 Etoposide 20
8.4 Methotrexate 21
8.5 Monoclonal Antibodies 21
8.6 Azathioprine and 6-Mercaptopurine 21
9. Heparin and Other Anticoagulants 21
9.1 Warfarin 22
9.2 Heparin (Unfractionated) 22
9.3 Direct-Acting Oral Anticoagulants (DOACs) 23
9.4 Thrombolytics 24
10. Safer Systems 24
10.1 Safe Administration of Oral, Enteral and Nebuliser Liquid Preparations 24
10.2 Standardisation of Terminology, Abbreviations and Symbols in the Prescribing and Administration of Medicines 24
10.3 Standardisation in Paediatric Prescribing 26
11. Antiarrhythmic Medications 26
12. Iloprost 26
13. Live Vaccines 27
14. Monoclonal Antibodies 27
15. Positive Inotropes and Vasopressors 28
16. High dose corticosteroid 28

Introduction

The purpose of the Sir Charles Gairdner Osborne Park Health Care Group (SCGOPHCG) High Risk Medication Policy and associated Register is to improve patient safety by raising awareness amongst all clinical staff who prescribe, dispense, handle or administer high risk medications at SCGOPHCG. High Risk Medications are medications that have an increased risk of causing significant patient harm or death if they are misused or used in error.

Errors with high-risk medications may not be more common than those from other groups but their consequences can be more devastating.

The purpose of identifying high risk medications is to establish locally based safeguards and strategies to reduce the risk of errors with these medicines during all phases of the medication use process.

This document does not replace the need for the application of clinical judgement to each individual presentation.

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☒
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☐
Other:	☐		

Definitions

High Risk Medication	A medication with an increased risk of causing significant patient harm or death when prescribed, administered or dispensed in error. Includes medications with a low therapeutic index and medications that present a high risk when administered via the wrong route or when other system errors occur.
High Risk Medication Register	Referred to in this document as ‘the Register’. A list of all medications considered to be High Risk at SCGOPHCG.
Medication Incident	A failure in the medication management cycle that leads to, or has the potential to lead to, harm to the patient. Errors rarely occur as the result of the actions of an individual, they are usually a result of a series of system failures. Incidents are classified under severity access codes (SAC) 1, 2 or 3.

Policy Principles

The SCGOPHCG High Risk Medication Register (See Appendix 1) will be determined, maintained, and periodically reviewed by the Medication Safety Committee and Drug and Therapeutics Committee.

The Register will contain information on the risks associated with each medication and the risk mitigation strategies currently in place to ensure safe use of each medication.

Clinical staff must be aware of medicines on the Register, and of the need to take extra care in their safe storage, handling, prescription, administration and monitoring with reference to local protocols.

If there is a high risk of death or serious injury to the patient where a medicine is inadvertently misused or administered incorrectly, the medicine is to be included in the Register and appropriate protocols or guidelines developed.

At a minimum, the following classes of high risk medications recommended by the Australian Commission for Safety and Quality in Healthcare, and associated with the 'APINCHS' acronym, are considered for inclusion in the Register:

A	Antimicrobials
P	Potassium and other electrolytes; Psychotropic medications
I	Insulin
N	Narcotics / Opioids; Neuromuscular blocking agents
C	Chemotherapeutic agents
H	Heparin and other anticoagulants
S	Safer Systems (e.g., safe administration of liquid medications, independent double checking, standardised order sets and medication charts)

This is not an exhaustive register of high-risk medications, all medications carry risk of adverse events if prescribed, administered, or dispensed inappropriately. This register will be reviewed on a twice-yearly basis and when needed by the SCGOPHCG DTC.

Accountabilities and Responsibilities

The following groups and positions have responsibilities or requirements associated with the implementation or monitoring of this policy:

- **Western Australian Drug Evaluation Panel (WADEP)**
  - At a state level, WADEP is responsible for assessing the risk profile of each medication before addition to the Statewide Medicines Formulary (SMF).

- **Chief Executive, Health Service Executives, Managers**
  - Ensure the mandatory policy and standards are implemented at all health facilities.
  - Facilitating completion of compliance audits and addressing compliance issues identified as a result.

- **Safety and Quality Managers**
  - Ensure systems are in place to:
    - Implement the mandatory high risk medicine policy and standards.
    - Monitor compliance with the high-risk medicine policy and standards.

- **Drug and Therapeutics Committee**
  - Ensure a local register of high-risk medicines is current and maintained.
  - Development and endorsement of policies, protocols and guidelines relating to high-risk medicines.
  - Receive and review reports on high-risk medicine incidents through Datix Clinical Incident Management System (CIMS), policy compliance rates and formulate corrective action, if indicated.

- **Clinical staff involved in medication management**
  - Comply with policy and standards for high-risk medicines.
  - Follow the local protocols/guidelines described below in the High-Risk Medication Register.
  - When prescribing and supplying high risk medicines, provide clear advice to ensure their safe administration.
  - Identify high risk medications on discharge and ensure adequate patient education is provided.
  - Maintain knowledge base relevant to area of practice.
  - Where appropriate, ensure that intravenous high-risk medications are administered via the B Braun ® Infusomat Pump using the SCGOPHCG Smart Infusion Pumps and Dose Error Reduction Software
  - Report all clinical incidents involving high risk medications via the clinical incident management system (Datix CIMS)

Compliance, monitoring and evaluation

Ensuring compliance against this policy is the responsibility of the Director of Clinical Services and will be evaluated by:

- Routine clinical incident review process
- Audit processes

Related Documents

- Western Australian Medicines and Poisons Act 2014
- Western Australian Medicines and Poisons Regulations 2016
- WA Department of Health MP 0077/18 Statewide Medicines Formulary Policy
- WA Department of Health MP 0078/18 Medication Chart Policy
- WA Department of Health MP 0131/20 High Risk Medication Policy
- WA Department of Health MP 0122/19 Clinical Incident Management Policy
- SCGOPHCG Medicines Management Policy
- SCGOPHCG Smart Infusion Pumps and Dose Error Reduction Software Policy
- NMHS Medicines Management

References

1. Australian Commission on Safety and Quality in Healthcare. High Risk Medicines. https://www.safetyandquality.gov.au/our-work/medication-safety/high-risk-medicines
2. Australian Commission on Safety and Quality in Healthcare. National Safety and Quality Healthcare Standards. Standard 4 – Medication Safety. Action 4.15: High-risk medicines https://www.safetyandquality.gov.au/standards/nsqhs-standards/medication-safety-standard/medication-management-processes/action-415
3. Fiona Stanley Hospital Fremantle Hospitals Group. South Metropolitan Health Service. High Risk Medications Policy. July 2021. Reference #:FSH-PHARM-POL-0005.
4. Institute for Safe Medication Practices (ISMP).ISMP List of High-Alert Medications in Acute Care Settings. ISMP; 2018. https://www.ismp.org/recommendations/high-alert-medications-acute-list
5. Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook, Australian Government Department of Health and Aged Care, Canberra, 2022, immunisationhandbook.health.gov.au
6. Royal Perth Bentley Group. East Metropolitan Health Service. High Risk Medication Policy. November 2020.
7. Western Australian Country Health Service. High Risk Medications Procedure. March 2022.

Authorisation & Version Control

Policy Sponsor	Director of Clinical Services				
Policy Contact	Medication Safety Pharmacist				
Date First Issued:	19/04/2023	Last Reviewed:	17/03/2023	Review Date:	17/03/2026
Version No.	1.1	Reference #:	HP261		
Approved by:	Director of Clinical Services			Date:	19/04/2023
Endorsed by:	SCGOPHCG Policy and Guideline Committee			Date:	19/04/2023
Standards Applicable					
Aboriginal Health ISD Number:	2205				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

Appendix 1. High Risk Medication Register

1. High Risk Medicines Management Strategies

- A culture of patient safety, based on the principles of just culture provides a solid foundation for safe and effective systems and teamwork. In a just culture, safety is valued, reporting of safety risks is encouraged without penalisation, and the staff, leadership and executive team are held accountable using a clear transparent process that evaluates errors.
- All medication incidents involving a high-risk medication that have caused moderate to severe harm (SAC 1 or SAC 2) must be investigated to determine improvements to practice to prevent future incidents.
  - This process is under the governance of the Clinical Risk Management Team, the SQP Medication Safety Pharmacist and other clinical staff as stakeholders.
  - Investigation of SAC 1 and SAC 2 incidents are conducted via panel review to implement systemwide changes where gaps are identified
  - All reports from investigations are sent to the DTC for noting

- The Medication Safety Committee will conduct regular reviews (monthly) of all medication incidents reported via Datix CIMS to identify trends in incidents.
- The availability of high-risk medicines on imprest should be limited to areas where there is a clinical need for them.
- Set mechanical devices (e.g., Drug Error Reduction Software) so that they default to the safety setting for the relevant clinical area.
  - SCGOPHCG Smart Infusion Pumps (SIPS) and Dose Error Reduction Software (DERS) Policy)

2. Antimicrobials

Specific antimicrobial agents that have a high risk of causing harm include aminoglycosides, amphotericin and vancomycin.

Refer to:

- SCGH Intravenous Antimicrobial Administration Guidelines for Inpatients
- SCGOPHCG Antimicrobial Stewardship (AMS) Policy
- SCGH Healthy Antimicrobial Prescribing Initiative (HAPI) Intranet Page
- SCGOPHCG Neutropenic Sepsis Guideline
- SCGH MMG Post Exposure Prophylaxis (PEP) HIV Pre-Packed Starter Packs
- SCGH MMG Home Link IV Antibiotic Infusion Rates
- Therapeutic Guidelines Australia
- WA Department of Health Antimicrobial Stewardship Guidance Document

2.1 Aminoglycosides (Gentamicin, Tobramycin and Amikacin)

Refer to:

- SCGH Drug Guideline - Gentamicin (not DTC-approved)
- SCGH Medication Information Leaflet – Aminoglycoside Antibiotics
- Therapeutic Guidelines Australia – Principles of Aminoglycoside Use

2.2 Amphotericin

Confusion between the formulations of amphotericin B:

- Amphotericin B (Fungizone® also referred to as amphotericin B deoxycholate),
- Amphotericin B liposomal (AmBisome®), and
- Amphotericin B phospholipid complex (Abelcet®)
  may result in both prescribing and administration errors.

Amphotericin B deoxycholate (Fungizone®) is only available through the Special Access Scheme (SAS) and stored separately to AmBisome®, the amphotericin B liposomal formulation.

Preparation of any sterile amphotericin product must be undertaken by trained Aseptic Services pharmacy staff due to specialised preparation requirements. Awareness of these different formulations and differing dosage recommendations helps to reduce the risk of under- or over-dosing.

Refer to:

- SCGH MMG Amphotericin B Intravenous – Liposomal (AmBisome®)
- WATAG WA Medication Safety Group Alert - Confusion between non-lipid and lipid formulations of injectable amphotericin

2.3 Vancomycin

Incorrect dosing with respect to weight and renal function can result in nephrotoxicity and ototoxicity.

Refer to:

- SCGH MMG Vancomycin IV Guideline

3. Potassium and Other Electrolytes

Potassium and certain other electrolytes may be administered intravenously to correct clinical deficiency in patients for whom oral electrolyte supplementation is not appropriate or when urgent replacement is required.

The use of pre-mixed preparations of electrolytes for intravenous administration is preferred, where they are available.

3.1 Potassium

Potassium salts are administered intravenously to correct clinical deficiency in patients for whom oral electrolyte supplementation is not appropriate or when urgent replacement is required.

Errors in the preparation and administration of intravenous potassium can be fatal.

Potassium is available as chloride, phosphate and acetate salts.

Storage of high concentration potassium solutions for intravenous administration (including potassium chloride, potassium dihydrogen phosphate and potassium acetate) are restricted to pharmacy and approved clinical areas where they are kept in a red container separate from other IV medications.

Pre-mixed potassium infusion bags are to be used in preference to preparing non-standard solutions in ALL clinical areas (including critical care areas).

The maximum infusion rate of potassium chloride via a peripheral line is 10mmol per hour. The only exceptions are in the management of Adult Diabetic Ketoacidosis and Adult Hyperglycaemic Hyperosmolar State where up to 20mmol per hour of potassium chloride can be administered peripherally.

For infusion rates > 10mmol per hour, it is recommended the solution be infused via a central line (CVC) under close nursing supervision.

Patients receiving intravenous potassium replacement at a rate in excess of 10mmol per hour require continuous cardiac monitoring and must be cared for in a high acuity area that can facilitate this.

Infusion rates >10mmol per hour are not approved at Osborne Park Hospital. Patients requiring these faster infusion rates must be transferred to SCGH.

Refer to:

- SCGH MMG Hyperkalaemia
- SCGH MMG Hypokalaemia
- SCGH MMG Potassium Supplements - Intravenous
- SCGH MMG Potassium Supplements - Oral
- SCGOPHCG Adult Diabetic Ketoacidosis (DKA) Guideline
- ACSQHC High Risk Medication Alert – Intravenous Potassium Chloride

3.1 Calcium

Calcium when administered intravenously can be rapidly fatal in overdose.

Calcium for intravenous administration is available as a calcium chloride or calcium gluconate salt. Calcium chloride is three times more potent than calcium gluconate.

If injected too rapidly this may cause peripheral vasodilation, bradycardia, hypotension, cardiac arrhythmias and cardiac arrest.

Calcium gluconate is a supersaturated solution and precipitation may occur. Vials should be thoroughly inspected before use and should not be used if the solution is discoloured, cloudy, turbid, or if a precipitate is present.

Solutions are highly irritant, and extravasation can cause severe complications.

Calcium levels should be monitored every 2 to 4 hours if given by continuous infusion.

Refer to:

- Peri-operative Management of Patients with Advanced Chronic Kidney Disease undergoing Parathyroidectomy
- Therapeutic Guidelines: Hypocalcaemia
- Australian Injectable Drugs Handbook

3.2 Magnesium

Magnesium for injection is available as a magnesium chloride or magnesium sulfate salt. They are not directly interchangeable.

Excessive administration can cause nausea, vomiting, hypotension, muscle weakness, muscle paralysis and CNS depression. Severe hypermagnesaemia may result in respiratory depression, muscle paralysis, coma, cardiac arrhythmias, and cardiac arrest.

Ensure an intravenous preparation of a calcium salt is available to reverse the effects of magnesium toxicity if required.

Refer to:

- SCGH MMG Hypomagnesaemia

3.3 Phosphate

Potentially fatal hyperkalaemia can develop rapidly and asymptomatically with use of the potassium dihydrogen phosphate and dipotassium hydrogen phosphate salts.

Phosphate is available as sodium or potassium salts. Refer to section 3.1 Potassium for additional information on safe administration of potassium salts.

Excessive IV phosphate administration may cause hyperphosphataemia, hypocalcaemia and arrhythmias.

Correct hypocalcaemia (if present) prior to replacing phosphate. Monitor serum sodium, potassium, phosphate and calcium concentrations as well as renal function every 12 to 24 hours.

Refer to:

- SCGH MMG Hypophosphataemia

3.4 Sodium Chloride – Hypertonic Saline

Hypertonic saline may be used to treat hyponatraemia. Caution is required when administering hypertonic saline due to risk of osmotic demyelination (which may produce permanent central nervous system injury and/or be fatal) if abnormalities in plasma sodium are corrected too rapidly.

Hypertonic saline solutions for nebulisation may be prepared for respiratory indications.

Storage of sodium chloride 23.4% vials is restricted to specific clinical areas.

Refer to:

- Therapeutic Guidelines: Hyponatraemia

4. Psychotropic Agents

Psychotropic agents (including antipsychotics, antidepressants, benzodiazepines and stimulants) carry certain risks.

While psychotropic agents generally do not carry high risks as a broad category, they raise risks in specific patient populations or clinical situations:

- Risk is increased when prescribed in combination or at high doses.
- Given the risk of self-harm and suicide is increased within cohorts of individuals with mental illness, psychotropics may pose an increased risk of overdose for those individuals who have risk factors (also noting the therapeutic impact of psychotropics can reduce suicide rates).
- Antipsychotic medications pose long term health risks which, in conjunction with associated inherent mental illness risk, may predispose patients to metabolic syndrome.

- Psychostimulants and sedative medication may pose a potential risk of diversion and misuse.
- The use of antipsychotics in patients with dementia is associated with an increased risk of cerebrovascular events (including stroke) and death due to any cause. If these medications are provided on discharge, a management plan for de-escalation should be documented in the medical record and discharge summary.

4.1 Clozapine

Clozapine requires strict monitoring in light of its potential to cause neutropenia, agranulocytosis, myocarditis, cardiomyopathy and gastrointestinal hypomotility.

Supply of clozapine is only by Pharmacy on an individual patient basis.

Refer to SCGH Clozapine Medication Management Procedure for the safe prescribing, administration, monitoring and supply of clozapine at SCGOPHCG.

Refer to:

- SCGH Clozapine Medication Management Procedure
- WA Clozapine Initiation and Titration Chart
- ClopineCentral
- Clozapine eLearning package
- Clozapine Monitoring Form A and Clozapine Monitoring Form B
- These forms are designed to document the face to face clinical review and monitoring requirements for consumers on clozapine. These non-mandatory forms can be used after the WA Clozapine Initiation and Titration Chart.
- NMHS Mental Health Policy – Clozapine Prescribing and Monitoring
- This document provides guidance on prescribing and monitoring of clozapine.
- Clinician prompt checklist to assess clozapine side effects
- This assessment checklist supports the assessment of potential side effects related to clozapine therapy.
- Side effects associated with clozapine therapy
- This table outlines the more common side effects related to clozapine.
- WA Clozapine Initiation and Titration Chart Education Resource
- This PowerPoint has been developed to provide education on features, prompts and alerts in the chart.
- WA Clozapine Initiation and Titration Chart User Guide

4.2 Lithium

Lithium is indicated for treatment of acute mania, acute bipolar depression, prophylaxis of bipolar affective disorder and as augmentation therapy in depression, schizoaffective disorders, and schizophrenia. It is available as immediate release and slow release preparations.

Lithium has a narrow therapeutic index and failure to recognise the early signs of toxicity may lead to delays in treatment, resulting in poor patient outcomes including, in the worst cases, death. Early symptoms of lithium toxicity are varied and non-specific. They are more likely to occur when serum lithium concentration exceeds 1.5 mmol/L but can occur when serum lithium levels are within the target concentration range.

- There are relatively narrow margins between therapeutic and toxic doses of lithium and therefore regular blood tests and clinical monitoring is important. In addition,

toxicity occurring close to or within target serum lithium concentration range is a known risk factor for poor outcomes.
- Regular blood tests to monitor renal function, thyroid function and lithium levels are recommended to ensure toxicity does not eventuate.

Main causes of toxicity:

- Interactions with other medicines that affect renal function (e.g. diuretics, NSAIDs, angiotensin converting enzyme (ACE) inhibitors, angiotensin II receptor blockers),
- Reduced fluid intake, fluid loss from vomiting, diarrhoea, excessive sweating or nephrogenic diabetes insipidus,
- Impaired renal function,
- Reduced salt intake,
- Advanced age,
- Concurrent illness,
- Chronic supratherapeutic dosing
- Deliberate or inadvertent overdose

Signs and symptoms of toxicity:

- Mild - blurred vision, nausea, vomiting, diarrhoea, drowsiness, muscle weakness, ataxia, apathy
- Severe - confusion, disorientation, unsteadiness, coarse tremor, increased muscle tone and myoclonic jerks acute renal failure

Refer to:
- NMHS Mental Health Policy – Lithium: Prescribing and Management
- WA Health Printable Choice and Medication© Leaflets

4.3 Zuclopenthixol Acetate

Zuclopenthixol acetate (Clopixol Acuphase®) is an intermediate acting intramuscular injection indicated for managing acute psychosis and mania in adults, particularly when a patient is hostile or aggressive.

It can lead to significant extrapyramidal side effects (EPSE) and drowsiness that persists for a prolonged period of time. Zuclopenthixol acetate must only be prescribed with the approval of a consultant psychiatrist.

Refer to:
- NMHS Mental Health Policy – Zuclopenthixol Acetate (Clopixol Acuphase®) Prescribing

5. Insulin

Insulin is a high-risk medication due to its most frequent and serious adverse effect; hypoglycaemia, which can manifest as sweating, hunger, faintness, palpitations, tremor and in severe cases, coma, and death.

Errors in insulin therapy can cause serious harm or can be fatal.

Confusion may arise between the different insulin preparations available, especially if not prescribed in a consistent and clear manner.

Intravenous insulin is lethal if given in substantially excessive doses or in place of other medications (insulin and heparin are often mistaken for one another since both are ordered in units).

Refer to:

- SCGOPHCG NPG Diabetes Patient Management
- OPH Insulin Infusion Pump Management

5.1 Insulin by Subcutaneous Injection

5.1.1 Prescribing Insulin by Subcutaneous Injection

Subcutaneous insulin should be prescribed on MR846/MR(OPH)156T Insulin Subcutaneous Order and Blood Glucose Record - Adult.

Care must be taken to ensure that the insulin type and administration instructions are fully documented including:

- Full trade name – e.g., Humulin 30/70 not just Humulin, Humalog Mix 50 not just Humalog Mix
- Device or product – cartridge/prefilled pen/vial
- Specific time of administration – immediately before meals or specific time to be given in respect to food
- Dose - available charts are pre-printed with the word 'UNITS' written in full to avoid confusion

Refer to:

- Australian Commission on Safety and Quality in Health Care - Recommendation for terminology, abbreviations and symbols used in medicines documentation

5.1.2 Administering Insulin by Subcutaneous Injection

- Cartridges, pen devices and vials are for individual patient use.
- The BD Autoshield™ safety pen needle with the pen injector device is to be used for all subcutaneous insulin administration by nursing staff and for patient self-administration at SCGOPHCG.

Refer to:

- SCGOPHCG NPG Diabetes Patient Management

5.1.3 High Concentration Insulin

A number of high concentration insulin preparations are available in Australia.

Toujeo® is approved for inclusion on the State Medicines Formulary (SMF) and can be used at SCGOPHCG in accordance with the SMF.

Refer to:

- WATAG Advisory Note – Safety Alert: High Concentration Insulin

5.1.3.1 Toujeo® (insulin glargine 300 units/mL)

Toujeo® is a high concentration insulin product that is three times the concentration of standard insulin preparations. Toujeo® is not directly interchangeable with other insulin glargine preparations.

Refer to:

- WA Medication Safety Alert - High Concentration Insulin - Toujeo®

5.2 Storage of Insulin

- Unopened vials/cartridges/prefilled pens should be stored in an approved medication fridge (at 2 to 8 degrees Celsius).
- Insulin in use can be stored at room temperature (below 25 degrees Celsius) for up to 28 days.

- Insulin vials must be for individual patient use only.
- When the insulin is used for the first time, ensure a label is affixed to note the date and time of opening and patient identification for whom the insulin is to be used.
- Ensure insulin is discarded if it has been out of the fridge for 28 days or more.
- Do not place insulin in, or close to the freezer compartment as it must not be frozen.
- Do not expose vials, cartridges or prefilled pens to sunlight or temperatures above 25 degrees Celsius.
- Do not use insulin if it has expired (always check the original packaging for the expiry date).

5.3 Administering Insulin by Intravenous Infusion

Intravenous insulin is lethal if given in substantially excessive doses or in place of other medications.

Insulin must be prescribed on one of the following insulin prescription and monitoring charts with full instructions:

- MR826/MR(OPH)156A Adult Variable Rate Intravenous Insulin Management Record
- MR849 Adult Diabetic Ketoacidosis (DKA) Guidelines and Management Record
- MR848 Adult Hyperosmolar Hyperglycaemic State Guidelines and Management Record

When administering intravenous insulin:

- Insulin infusions should be administered via a programmable pump, ideally a pump that utilises Drug Error Reduction Software (DERS).
- Double check the concentration and the infusion rate are consistent with the prescription to ensure the correct dilution is used.
- Problems may arise if pumps are programmed incorrectly.
- Insulin infusions must never be run unopposed and are to be administered alongside a concurrent a glucose infusion.
- Hypoglycaemia is a well-recognised complication of insulin treatment, especially when prescribed for the management of hyperkalaemia. Refer to the SCGH MMG Hyperkalaemia for guidance.

Refer to:

- SCGH MMG Hyperkalaemia
- SCGOPHCG Insulin: Adult Variable Rate Intravenous Guideline
- SCGH Adult Diabetic Ketoacidosis (DKA) Guideline
- SCGOPHCG Adult Hyperosmolar Hyperglycaemic State (HSS) Guideline
- SCGOPHCG NPG Intravenous Therapy

6. Narcotics/Opioids and Sedative Agents

Opioids are a high-risk medication. Risks include:

- respiratory depression (opioid induced ventilatory impairment),
- increased falls risk,
- increased risk of motor vehicle accident, especially when taken concomitantly with other CNS depressants
- inadvertent initiation of persistent opioid use,
- non-medical use of prescription opioids, in addition to
- the commonly recognised side-effects of constipation, nausea and vomiting.

These effects can be increased by other medications and alcohol consumption.

SCGOPHCG has an Acute Pain Service, a Chronic Pain Service and a Palliative Care Service. Refer to HealthPoint for further information and contact details for these services.

Refer to:

- SCGH MMG Community Program for Opioid Pharmacotherapy (CPOP)
- OPH Community Program for Opioid Pharmacotherapy (CPOP)
- SCGH MMG Intranasal Fentanyl
- SCGH MMG Oral Opioid Prescribing in Acute Non-Cancer Pain
- SCGH Intravenous Analgesia Protocols (Schedule 8) in PACU
- SCGOPHCG Discharge Analgesia Prescribing Guideline
- SCGOPHCG NPG Opioid and Ketamine Intravenous Analgesia
- SCGOPHCG NPG Regional Analgesia (Acute Pain
- SCGOPHCG Medicines Management Policy
- OPH NPG Drug Administration
- OPH Surgical Services Drug Administration in Theatre Guideline
- NMHS Administration of S8 and S4R Medicines Policy and Procedure
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicines Policy
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicines Procedure
- WA Medication Safety Alert – Gabapentinoids in combination with opioids and the risk of respiratory depression
- WA Community Program for Opioid Pharmacotherapy (CPOP)
- WATAG – Guidelines for the Pharmacological Treatment of Neuropathic Pain
- WATAG – Recommendations for prescribing analgesia on discharge following surgery or acute injury
- Western Australian Medicines and Poisons Act 2014
- Western Australian Medicines and Poisons Regulations 2016
- WA Department of Health MP 139/20 Medicines Handling Policy

6.1 Prescribing and Administering Opioids and Sedative Agents

When prescribing analgesia, including opioids and other adjunctive agents, consider:

- Patient's clinical status, including opioid status (naïve vs tolerant)
- Age
- Renal and hepatic function
- Co-morbidities
- Other sedative medications contributing to respiratory depression and falls risk
- Potential drug interactions

Extra care should be taken when considering an opioid rotation. Independently check doses and opioid conversion tables prior to prescribing and administration where possible. Monitor patient to assess pain level, sedation and respiratory rate during this period. Refer to the WA Cancer and Palliative Care Services Opioid Conversion Guide for further information on opioid conversion.

Parenteral analgesia must be prescribed on the relevant prescription and monitoring chart with full instructions:

- MR810 Intravenous Analgesia Chart
- 810/MR(OPH)177B Regional Analgesia Chart
- MR(OPH)177A PCIA/Intravenous Opioid Infusion/NCA Standard Orders
- MR815 Prescription & Monitoring Form Subcutaneous Syringe Driver
- SCGH410/805 OPH170.1 WA Hospital Medication Chart – Short Stay

- SCGH401/805 OPH170.2 WA Hospital Medication Chart – Long Stay

The risk of inadvertent overdose, or over-sedation or respiratory depression from concurrent prescription of other sedative opioid drugs must be considered in patients who have been administered intrathecal or epidural opioids peri-operatively.

Naloxone injection should be available in all clinical areas for opioid reversal.

Regularly review patients using an appropriate pain scoring tool to assess response to analgesia.

Monitoring of sedation levels and respiratory rate should also be performed regularly.

6.2 Patient Controlled Intravenous Analgesia

A prescription should only be initiated and altered by:

- A medical member of the Acute Pain Service
- An Anaesthetist
- An Intensive Care or Emergency Medicine Practitioner in consultation with an APS medical practitioner.

No additional opioids or sedatives are to be administered unless authorised by the PCIA prescriber, due the risk of inadvertent overdose or respiratory depression

Refer to:

- SCGOPHCG NPG Opioid and Ketamine Intravenous Analgesia
- SCGOPHCG NPG Regional Analgesia Acute Pain

6.3 Narcotic Transdermal Patch Delivery Systems

A number of medications are available in a patch formulation including potent opioid analgesics (e.g. fentanyl and buprenorphine). It is important to ascertain the presence of any transdermal patches at time of admission.

- Absorption of the drug from a patch can be increased by exposing the patch to heat sources such as heat packs, electric blankets and hot baths. A high fever can also increase absorption.
- Record time of application and time of removal on medication chart. Document site of application of the patch in the patient's medical record.
- A transdermal patch check sticker is available to support documentation of nursing checks of prescribed patches in situ.
- Ensure that the transdermal patch has been removed if the prescription order has been ceased.
- Ensure that the previous patch has been removed before applying a new patch. Both the application and removal of patch should be recorded on the medication chart.
- Transdermal opioid patches should not be used in opioid-naïve patients unless specialist advice has been provided by appropriately trained medical staff.
- A significant amount of drug remains in a patch after its intended application period has expired. Used transdermal patches should be handled wearing gloves to avoid inadvertent contact of patch with skin, folded onto itself so the adhesive sides stick together, cut into small pieces with scissors and placed into a pharmaceutical/clinical waste bin for incineration bin.

Refer to:

- Guidelines for the WA Hospital Medication Chart (Adult and Paediatric) – Transdermal Patch Sticker

- SCGOPHCG Medicines Management Policy
- NMHS Administration of S8 and S4R Medicines Policy and Procedure
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicines Policy
- NMHS Destruction, Discard and Disposal of S8 and S4R Medicines Procedure

6.4 Restricted Schedule 4 (S4R) Medicines

Restricted Schedule 4 Medicines are a range of Schedule 4 medicines that are liable to abuse, misuse and diversion. Additional controls around storage and record keeping are required within the public hospital system.

Refer to:

- SCGH MMP Propofol Storage and Recording
- OPH Drug Administration in Theatre Guideline
- SCGH MMP Patient's Own Medicines
- SCGH MMP Supply of S4Rs and S8s from Pharmacy to Wards on Weekends/Public Holidays
- SCGH NPG Medication Management

7. Neuromuscular Blocking Agents

Examples of neuromuscular blocking agents (NMBAs) include:

- atracurium,
- cisatracurium.
- mivacurium,
- pancuronium,
- rocuronium,
- suxamethonium,
- vecuronium,

NMBAs are considered high risk medications because inadvertent use in patients without the availability of medical staff skilled in airway support can lead to respiratory arrest, permanent harm and death. Serious incidents have occurred involving inadvertent administration of NMBAs to a patient instead of another agent (e.g. sedative).

These medications are used during tracheal intubation, during surgery of intubated patients, and to facilitate mechanical ventilation of critically ill patients.

Administration errors involving these medications in Australia and internationally have been associated with look-a-like packaging and labelling; and look-a-like/sound-a-like medication names; resulting in selection errors. These risks can be exacerbated when NMBA are stored with other medications. It is recommended that NMBAs are segregated and sequestered to differentiate all neuromuscular blockers from other medications.

Refer to:

- SCGH Critical Care Guideline – Cisatracurium in Acute Respiratory Distress Syndrome (not DTC-approved)
- SCGH ICU Standard Intravenous Drug Dilution, Dose and Special Precautions (not DTC-approved)

7.1 Storage Requirements for NMBAs

NMBAs should only be stored in areas that routinely use these medications. Approval to store NMBAs on imprest is required from the SCGOPHCG Drugs and Therapeutics Committee. The storage location of all NMBAs is reviewed annually by the SCGOPHCG DTC.

In areas permitted to store NMBAs, they must be stored in a clearly marked and red lidded container.

An exemption to the requirement to store NMBAs in a clearly marked and sealed container has been granted by the SCGOPHCG DTC in the following locations:

- Anaesthetic trolleys
- MET bag/kit, Transport bag

Segregate NMBAs from all other medications in the pharmacy department by placing them in separate red lidded containers in the refrigerator or other secure, isolated storage area.

Auxiliary labels should be placed on all storage containers and drawers that contain NMBAs that state "WARNING: Paralysing agent. Causes respiratory arrest – patient must be ventilated".

7.2 Prescribing Requirements for NMBAs

- Outside the operating theatre or procedural areas, orders for NMBAs should only be part of an intubation protocol to maintain a specific level of paralysis while the patient is mechanically ventilated.
- Prescribing of NMBAs is restricted to clinicians who have adequate knowledge and training such as an Anaesthetist, Emergency Department Physician or Intensivist.

7.3 Preparation and Administration of NMBAs

- Proper labelling of all syringes containing NMBAs must comply with the National Standards for User-applied Labelling of Injectable Medicines, Fluids and Lines (the Labelling Standards). (See section 10.2.1)
- Ensure all appropriate reversal agents for neuromuscular blockade are available to qualified staff that might need them in an emergency.
- NMBAs should only be administered by staff with experience in maintaining an adequate airway and respiratory support, and only in units where intubation and respiratory support can be provided.

All chemotherapeutic agents are considered high risk medications.

8. Chemotherapeutic Agents

- All staff involved in the handling of chemotherapeutic agents must have the relevant knowledge and skills and be competent to perform the tasks required of their role, which may include prescribing, dispensing, preparation or administration of chemotherapeutic agents.
- A cytotoxic warning label must be placed on any dispensed chemotherapy, antineoplastic and cytotoxic agents and detailed via appropriate annotation of the medication administration chart (if not MOSAIQ) to identify this risk associated with handling.
- All prescriptions for intermittent therapy should have clearly documented start and stop dates.
- All treatments should be clinically verified by a pharmacist prior to dispensing. The pharmacist should have access to the patient information relevant to the treatment.
- All prescriptions for chemotherapy should be clinically verified by a chemotherapy competent pharmacist before it is dispensed in the Aseptic Services Unit and by a chemotherapy competent registered nurse prior to administering the dose.

- All chemotherapy and associated therapy should be clinically verified by a nurse and the therapy checked against the order by 2 appropriately trained and credentialed health professionals prior to administration.
- Patients should be given both written and verbal information about their treatment to include all medications, expected side effects, how to take supportive medication and who to contact in the event of an emergency or severe adverse events.

Refer to:

- eviQ Cancer Treatments Online
- COSA guidelines for the safe prescribing, dispensing and administration of systemic cancer therapy. Clinical Oncology Society of Australia. August 2017.
- SCGH Electronic Chemotherapy Manual (ECheMa)
- SCGH MMP MOSAIQ Care Plan Governance
- SCGOPHCG Intravesical Chemotherapy Guideline
- SCGOPHCG NPG Antineoplastic Hazardous Agent Management
- OPH Pharmacy Procedure Cytotoxic Drug Therapy & Safe Handling of Cytotoxic Drugs

8.1 Vinca Alkaloids

Vinca alkaloids are a class of chemotherapeutic agents which includes vinblastine, vincristine, vinflunine and vinorelbine.
While any medication administered via the incorrect route can result in harm, the inadvertent intrathecal administration of vinca alkaloids has resulted in permanent disability or death and remains a potential risk.

Vinca alkaloids can be fatal if given by the intrathecal route and must ONLY be administered intravenously.

Vincristine in particular has been involved worldwide in a large number of fatalities from inadvertent intrathecal administration. The guidance below is designed to eliminate opportunities for such administration error.

Refer to:

- ACSQHC High Risk Medication Alert - Vincristine
- WA Health Mandatory Standard for vinca alkaloids

8.1.1 Safe Prescribing of Vinca Alkaloids

In accordance with the Clinical Oncology Society of Australia guidelines, vinca alkaloids must only be prescribed by medical practitioners with appropriate skills, training and qualifications in the management of cancer.

8.1.2 Preparation and Dispensing of Vinca Alkaloids

- Vinca alkaloids must be prepared and supplied in an infusion bag or CADD device containing a total volume of at least 50mL; *never* in a syringe.
- Vinca alkaloids must only be prepared and dispensed by appropriately trained staff who have been assessed as competent to prepare and dispense chemotherapy.
- All vinca alkaloids should be labelled clearly with the warning ‘FOR INTRAVENOUS USE ONLY – FATAL IF ADMINISTERED BY ANY OTHER ROUTE’.

8.1.3 Administration of Vinca Alkaloids

- Vinca alkaloids must never be administered intrathecally.
- Vinca alkaloids must only be administered by appropriately trained staff who

have been assessed as competent to administer chemotherapy.
- Staff administering vinca alkaloids must be aware of the risk of extravasation and ensure procedures for preventing, monitoring for, and treating extravasation are followed.
- Vinca alkaloids must never be administered using a motorised pump through a peripherally inserted intravenous cannula. They should be administered using gravity to reduce the risk of extravasation.

8.2 Intrathecal Chemotherapy

The consequence of administering chemotherapeutic agents not intended for intrathecal use via this route can be fatal.

Refer to:
- SCGOPHCG NPG Intrathecal Drug Delivery
- SCGH Haematology Department Manual – Intrathecal Chemotherapy Policy (not DTC-approved)
- SCGH Intrathecal Chemotherapy e-Learning Package

8.2.1 Preparation, Dispensing and Delivery of Intrathecal Chemotherapy

- All intrathecal injections must be placed inside a sealed yellow outer bag.
- All layers of outer packaging including the outer sealed bag must be labelled with the patient name, name of drug and a clear warning “For Intrathecal Use Only”.
- Syringes for intrathecal therapy must be labelled with a warning “For Intrathecal Use Only”. To avoid obscuring information, “flag”-type labels must be applied to small syringes.
- Intrathecal injections prepared by the Pharmacy Department shall be delivered to the designated location under the direct control of an intrathecal-qualified nurse or pharmacist.
- In the case of the administration not preceding (e.g. failed lumbar puncture) the drug must be returned to the Pharmacy Department intact and safely disposed of. If another attempt to administer the drug is required, a new dose should be prepared.
- Where circumstances make same-day administration of chemotherapy drugs by intrathecal and intravenous routes unavoidable or dictated by protocol, the order of route of administration must be standardised, must be delivered separately in clearly demarcated containers and where possible the second drug to be administered shall be issued from pharmacy only after the pharmacist in charge has positively confirmed that the first therapy has been given successfully by examining the signed administration order.

8.3 Etoposide

- Etoposide is available as the base drug etoposide and the phosphate salt (Etopophos®). Both are available at SCGH depending on patient requirements and product stability.
- Etoposide is considered a high risk medication due to cases reported in the literature of confusion in dosing between etoposide and the pro-drug etoposide phosphate, resulting in significant harm to patients.
- All medication orders are prescribed, dispensed and otherwise represented at SCGH in the form ‘Etoposide (as phosphate) xmg’ or ‘Etoposide xmg’.
- Refer to MOSAIQ and ECheMa for protocols related to the prescription and administration of these agents at SCGH.

8.4 Methotrexate

- Oral methotrexate is used in the treatment of autoimmune or inflammatory disorders such as rheumatoid arthritis and severe psoriasis. Methotrexate is also used in the treatment of malignancies as part of specialised protocols.
- Australian cases with fatal consequences have been reported when oral methotrexate has been prescribed and administered more frequently than once weekly for autoimmune or inflammatory disorders.

Refer to:

- SCGH MMG Methotrexate Oral
- ISMP Patient Information Leaflet – Methotrexate

8.5 Monoclonal Antibodies

Refer to Section 14 Monoclonal Antibodies.

8.6 Azathioprine and 6-Mercaptopurine

- Immunomodulatory thiopurines, (azathioprine (AZA) and 6-mercaptopurine (6-MP or mercaptopurine)), are employed in the treatment of many conditions (including cancer and other non-cancer conditions). Their use is often limited by concerns of toxicity.
- The toxicity of azathioprine and 6-mercaptopurine is related to their metabolites in particular 6-thioguanine (6-GP) and 6–methylmercaptopurine (6-MMP).
- The relative accumulation of these metabolites depends on genetic polymorphism of the individual enzymes involved in their metabolism. A thiopurine methyltransferase (TPMT) activity level must be considered prior to initiation of therapy.
- Consideration should be made if a patient is prescribed other medications that may cause myelosuppression, or other medications that affect metabolism of AZA and 6-MP. Clinicians should be aware of the following drug interactions:
  - Allopurinol - reduces the metabolism of AZA/6-MP therefore increasing the risk of severe bone marrow toxicity. Avoid combination or reduce the dose to one-quarter of recommended dose
  - Febuxostat - use is not recommended in patients concomitantly treated with AZA/6-MP
  - Other medicines that can contribute to haematological toxicities e.g. clozapine, trimethoprim, etc.

Refer to:

- SCGH Medication Information Leaflet - Azathioprine

9. Heparin and Other Anticoagulants

There is potential for excessive bleeding with warfarin, heparin and other anticoagulants. The incorrect dose or failure to monitor therapy can contribute to this event. The patient's bleeding risk should be considered and documented on the Anticoagulation Medication Chart before commencing anticoagulant therapy and the VTE risk assessment must be completed on the Hospital Medication Chart. It is recommended that prescribers contact the Haematology team for advice on dosing if required.

All anticoagulants must be prescribed on the WA Anticoagulation Medication Chart. The WA Hospital Medication Chart must be annotated (cross-referenced) to identify when the WA Anticoagulation Medication Chart is in use to reduce the risk of duplicated orders or dose omissions. The WA Anticoagulation Medication Chart must be kept together with the current active WA Hospital Medication Chart.

Refer to:

- SCGOPHCG Periprocedural Management of the Anticoagulants Guideline
- SCGH Emergency Department Deep Vein Thrombosis Guideline (not DTC-approved)
- WA Department of Health MP 0078/18 Medication Chart Policy
- WA Anticoagulation Medication Chart and supporting resources

Two brands of warfarin (Marevan® and Coumadin®) are currently marketed in Australia. These brands are not bioequivalent and are available in different tablet strengths and colours.

9.1 Warfarin

Warfarin has the potential to interact with many medications that can alter the International Normalised Ratio (INR). Be aware that changing concurrent medications (by ceasing or adding a medication) may result in changes to INR and the dose of warfarin may require adjustment.

All new patients commencing warfarin therapy at SCGH should:

- Be referred to SCGH Home Anticoagulation Support Service AND
- Receive Marevan® for initiation as per the Statewide Medicines Formulary
  - A preference has been assigned due to the potential for serious medication errors resulting in unstable INR control and the need to protect the safety and wellbeing of patients.

Coumadin® should be prescribed and supplied only to patients already stabilised on Coumadin® therapy prior to admission.

Substituting or switching from Coumadin® to Marevan® in stabilised patients requires close monitoring.

Refer to:

- SCGH Emergency Department – Deep Vein Thrombosis Guideline (not DTC-approved)
- SCGH Perioperative Warfarin Management (not DTC-approved)
- Living with Warfarin: information for patients
- WATAG Guidelines for Anticoagulation using Warfarin

9.2 Heparin (Unfractionated)

- Prescribers should ensure the word “UNITS” is written in full to avoid confusion and errors of administration.
- Nurses should double check the correct number of units has been added to the syringe for injection or infusion and the infusion rate is consistent with the prescription to ensure the correct dose is administered to the patient.
- It is important to ensure that the correct dilution for heparin infusions is selected. It should be noted that there are two available dilutions for heparin infusions:
  - The standard dilution is 25,000 units in 500mL sodium chloride 0.9%.
  - If appropriate the fluid restricted patient dilution should be used - 25,000 units in 50mL. This is 10 times the difference to the standard nomogram in the WA Anticoagulation Medication Chart and can result in rate errors.
  - There is a separate low volume heparin infusion -nomogram that is available that outlines dilution and dosing for fluid restricted patients. When in use this should be clearly identified and attached to the patients WA Anticoagulation Medication Chart.

  - The application of the low volume heparin dilution is restricted and should only be used in specific clinical areas (ED, ICU, General HDU, CCU, Theatre, G63).
- The activated partial thromboplastin time (aPTT) has been used most widely for monitoring of therapeutic doses of unfractionated heparin in venous thromboembolism. A target ratio versus midpoint of normal range of 1.5-2.5 is employed.
- Standardisation between laboratories in the control of heparin therapy using the aPTT has not been achieved across all hospitals. This should be considered when interpreting results from patients transferred from another site.
- Platelet counts should be monitored every one to two days when prescribing heparin therapy. Heparins can cause thrombocytopenia which does not appear to be dose-related.
  - Heparin Induced Thrombocytopaenia (HIT) is an antibody-mediated adverse drug reaction that can lead to devastating thromboembolic complications, including pulmonary embolism, ischaemic limb necrosis, acute MI, and stoke.

Refer to:

- SCGH Emergency Department Deep Vein Thrombosis Guideline (not DTC-approved)
- SCGOPHCG MMG Heparin Induced Thrombocytopenia (HIT)
- SCGH Low Dose Heparin Guideline – Vascular Surgery
- OPH Venous Thromboembolism Prophylaxis Policy
- User Guidelines for the WA Anticoagulation Medication Chart

9.2.1 Low Molecular Weight Heparins (LMWH)

- Dose adjustment according to renal function and platelet count must be considered when prescribing low molecular weight heparins
- A baseline renal function test and full blood count should be done before commencing a LMWH. Dosing is weight based and must be modified in patients with renal insufficiency (creatinine clearance ≤ 30mL/minute).
- If therapeutic drug monitoring of LMWH is undertaken, it is recommended that an anti-Xa chromogenic assay is used. Samples taken at 4 to 6 hours after subcutaneous administration are suitable for assessment of peak anti-Xa level.

Refer to:

- SCGH Cardiovascular Medicine Area Specific Guideline - Clexane (Enoxaparin Sodium)(not DTC-approved)

9.3 Direct-Acting Oral Anticoagulants (DOACs)

Direct-acting oral anticoagulants include rivaroxaban, dabigatran and apixaban.

- Apixaban and rivaroxaban have no specific reversal agent.
- Idarucizumab (Praxbind®) is a specific reversal agent for dabigatran that is available at SCGH under direction of a Haematologist.
- Care is required when selecting patients for newer anticoagulant treatment and the following must be considered:
  - Dose adjustment is required in renal impairment (Creatinine Clearance ≤ 50mL/minute).
  - Use with caution in the elderly (> 75 years), underweight (< 50 kg) and overweight (>150kg)
  - Check for drug interactions and use of other anticoagulants, antiplatelet agents and NSAIDS.
  - Check for any existing co-morbidities with high risk of bleeding.

Plan ahead for surgery and other procedures. All new patients commencing on Venous Thromboembolism (VTE) DOAC therapy at SCGH should be referred to SCGH Hospital

Anticoagulation Support Services.

Prescribe on the WA Anticoagulation Medication Chart in the regular dose orders section to reduce the risk of unintended concurrent anticoagulant prescribing.

Refer to:
- WA Anticoagulation Medication Chart and supporting resources
- NSW Clinical Excellence Commission NOAC Guidelines
- Living with a direct-acting oral anticoagulant (DOAC) – patient booklet

9.4 Thrombolytics

Thrombolytics are associated with a potential for serious adverse effects including intracerebral haemorrhage, other serious bleeding events and allergic reactions. Care must be taken in screening patients for their use and close monitoring must be undertaken.

Refer to:
- Evidence-based resources and local guidelines.
- SCGH Cardiovascular Medicine Area Specific Guideline –Tenecteplase

WA Department of Health Protocol for Intravenous Thrombolysis in Acute Ischaemic Stroke

10. Safer Systems

10.1 Safe Administration of Oral, Enteral and Nebuliser Liquid Preparations

There is an identified risk of serious 'wrong route' medication errors resulting from accidental parenteral administration of solutions intended for oral, enteral or nebulised administration. Incidents involving oral, enteral and nebuliser solutions incorrectly administered parenterally have been reported nationally and internationally.

Preparations intended for oral, enteral or inhaled administration which have been drawn up into parenteral luer-lock syringes and inadequately labelled have resulted in clinical incidents, specifically errors of administration via the wrong route.

Oral/enteral non luer-lock syringes should be available in all clinical areas and used for all oral/enteral liquid preparations. Enteral syringes and giving sets should be labelled with a 'For Enteral Use Only' sticker.

Single-use nebules should be used wherever possible to minimise need to draw solutions into a syringe prior to administration. If it is necessary, an oral/enteral non luer-lock syringe can be used to draw up the medication. The syringe must be labelled with the intended route of administration with a 'For Inhalation Use Only' sticker.

Administration of medications intended for oral, enteral or inhalation use, when given by the parenteral route, results in a rapid absorption of the therapeutic agent into the blood stream. This can lead to a heightened risk of catastrophic outcomes which are often difficult to reverse.

Refer to:
- SCGOPHCG NPG Enteral Tubes
- SCGOPHCG NPG Nebulisation

10.2 Standardisation of Terminology, Abbreviations and Symbols in the Prescribing and Administration of Medicines

The use of inconsistent, ambiguous or non-standard abbreviations and terminology in the prescribing of medicines is a major cause of medication errors. Standardisation of terminology and abbreviations has been identified as an important strategy in reducing these errors.

Wherever possible, SCGH adopts the '[Recommendations for terminology, abbreviations and symbols used in medicines documentation'] (the Recommendations) developed by the Australian Commission on Safety and Quality in Health Care which provide:

- A standard for safe, clear and consistent terminology for medicines
- A list of safe terms, abbreviations and dose designations for medicines – summary sheet.

The Recommendations are applicable to all medication orders or prescriptions that are handwritten, pre-printed, computer generated (printed hard copy) or electronic, and all communications and records concerning medicines including telephone, verbal orders, prescriptions, medication administration records and labels for drug storage.

There may be specific circumstances where other terminology may be considered safe. However before these are included in local policies, the standards and principles outlined in the Recommendations should be applied.

Refer to:

- WA Department of Health WATAG Acceptable Prescribing Terms and Abbreviations

10.2.1 User Applied Labelling of Injectable Medicines, Fluids and Lines

Labelling is a recognised risk in the safe administration of injectable medications. Preparation of injectable medications for bolus injections or infusion is complicated with multiple opportunities for error. Labelling of injectable medications is often not done or incomplete, omitting information such as name of medication, dose, patient name or time of preparation.

It has been shown that errors in injectable medication administration are less likely to occur when a single person is responsible for preparing and labelling each injectable medication and that medication in well labelled syringes are more likely to have been prepared correctly.

- SCGOPHCG adopts the '*National Standard for User-applied Labelling of Injectable Medicines, Fluids and Lines*' (the Labelling Standards) which:
  - Assist health care professionals to identify the correct medicine and/or fluid and its correct administration/route/line at all times
  - Set out requirements for label inclusions and label placement
- The Labelling Standards are based on the following practice principles:
  - All medicines and fluids removed from the manufacturer's or hospital pharmacy's original packaging must be identifiable
  - All containers (e.g. bags, syringes) containing medicines leaving the hands of the person preparing the medicine must be labelled
  - Only one medicine should be prepared at a time and labelled before the preparation and labelling of a subsequent medicine

Any medicine or fluid that cannot be identified (e.g.an unlabelled syringe or other container) is considered unsafe and should be discarded.

10.3 Standardisation in Paediatric Prescribing

On occasion, children may be admitted as inpatients to SCGOPHCG for various reasons. In these instances, the following standards should be adopted in these patients.

Children are more prone to medication errors and are more vulnerable to harm from the effects of medication errors than adults. Dose calculation errors are one of the most common types of medication errors in children. Documenting patient weight and age, and adopting good prescribing, dispensing and administration practices can prevent patient harm associated with dosing errors.

The Paediatric National Inpatient Medication Chart (NIMC) must be used for children aged 12 years and under. The chart may be used in children up to the age of 18 years if deemed appropriate, particularly where weight-based dosing is being used. It includes additional features to improve prescribing safety for paediatric inpatients including:

- Dose calculation space for medicine orders
- Need for double signing when recording administration to document double-checking
- Additional spaces for weight, height, and the date these were recorded and gestational age at birth

Refer to:
- Information on the WA Paediatric NIMC

11. Antiarrhythmic Medications

Antiarrhythmic medications are associated with potential for serious adverse side effects, including proarrhythmic potential and a number of potentially serious drug interactions. Careful monitoring is required particularly with IV use of antiarrhythmic drugs.

Refer to:
- SCGH Cardiovascular Medicine Area Specific Guidelines – CCU IV Drugs (not DTC-approved)
- SCGOPHCG MMG Amiodarone
- SCGH MMG Digoxin
- SCGH Syringe Pump Infusions: Standard Dilutions in Critical Care
- SCGOPHCG NPG Intravenous Therapy

12. Iloprost

Iloprost is a prostacyclin analogue which is used in the treatment of Raynaud's phenomenon, scleroderma and critical limb ischaemia.

Inappropriate use of iloprost may be associated with hypotension, cerebrovascular ischemia/accident, myocardial infarction, arrhythmia, deep vein thrombosis.

Iloprost must only be prescribed under the direction of a Rheumatologist, Immunologist, Vascular Surgeon or Plastics Physician.

Refer to:
- SCGH Medication Information Leaflet - Iloprost
- SCGH MMG Iloprost IV

13. Live Vaccines

Most live vaccines are contraindicated in immunosuppressed patients due to the risk of infection from the vaccine virus. Serious complications may arise (including patient death). Any consideration to vaccinate in this patient cohort requires specialist review and input.

Live vaccines include:

- *BCG* (bacille Calmette–Guérin) vaccine
- Japanese encephalitis vaccine
- MMR *(*measles-mumps-rubella) vaccine
- Rotavirus vaccine
- Oral typhoid vaccine
- Varicella vaccine
- Yellow fever vaccine
- Zoster vaccine (Zostavax)

It is important to carefully review patients who are potentially immunocompromised for their suitability to receive a live vaccine. Live vaccines are generally contraindicated for most people who are severely immunocompromised.

Examples of patients who are immunocompromised include those who:

- have active leukaemia or lymphoma, or other generalised malignancy
- have received recent chemotherapy or radiotherapy
- have HIV (certain people only)
- had a solid organ transplant or haematopoietic stem cell transplant less than 2 years ago, or are still immunocompromised or taking immunosuppressive drugs, or have graft-versus-host disease
- are taking highly immunosuppressive therapy, including DMARDs or tsDMARDS (biological or targeted synthetic disease-modifying anti-rheumatic drugs), or high-dose corticosteroids
- have certain autoimmune diseases, particularly if they are on highly immunosuppressive therapy
- have aplastic anaemia
- have congenital immunodeficiency

Refer to:

- Australian Immunisation Handbook
- SCGH Haematology Policy Manual – Autologous bone marrow transplantation and stem cell mobilisation vaccination schedule (not DTC-approved)

14. Monoclonal Antibodies

Monoclonal antibodies are utilised for anti-cancer and non-cancer immunotherapy. When utilised as part of a chemotherapy regimen, treatment should be prescribed on the basis of a documented and referenced protocol and a treatment plan documented as per Section 8. Chemotherapeutic Agents.

Refer to:

- SCGH MMG Infliximab & Infliximab Biosimilars
- SCGH Medical Practice Guideline - Management of Immune Related Adverse Events Associated with Use of Immune Checkpoint Inhibitors

- SCGH MMG Monoclonal Antibodies: Safe Handling and Administration
- SCGH MMG Rituximab and Rituximab Biosimilars
- SCGH MMG Tocilizumab for Cytokine Release Syndrome
- SCGOPHCG NPG Intravenous Therapy
- SCGOPHCG NPG Antineoplastic/Hazardous Agent Management (in the Cancer and Non-Cancer Setting)
- SCGH Rituximab and Rituximab Biosimilars Immunosuppressive Therapy Chart

15. Positive Inotropes and Vasopressors

Vasoactive drugs are utilised in the critical care setting for management of patients with cardiovascular and/or respiratory instability.

Failure to manage these drugs appropriately can lead to haemodynamic instability, extravasation and, in extreme cases, death.

Restricted inotropes should be used in critical care and high acuity areas only as patients require continuous monitoring under the supervision of highly skilled staff.

Vasopressors have the potential to cause a number of significant complications, including hyponatraemia, dysrhythmias, myocardial ischaemia and ischaemic complications.

Additionally, a number of significant drug interactions exist with these medications.

Diligent use and careful monitoring is required.

Refer to:

- SCGH Cardiovascular Medicine Area Specific Guidelines – CCU IV Drugs (not DTC-approved)
- SCGH Cardiovascular Medicine Area Specific Guideline – Intravenous Levosimendan Administration (not DTC-approved)
- SCGH Critical Care Guideline - Adrenaline (not DTC-approved)
- SCGH Critical Care Guideline – Dobutamine (not DTC-approved)
- SCGH Critical Care Guideline – Milrinone (not DTC-approved)
- SCGH Critical Care Guideline - Vasopressin (not DTC-approved)
- SCGH ICU Standard Intravenous Drug Dilution, Dose and Special Precautions (not DTC-approved)
- SCGH MMG Dopamine
- SCGH MMG Metaraminol
- SCGOPHCG MMG Noradrenaline (for Central Line Administration)
- SCGOPHCG MMG Noradrenaline (for Peripheral Line Administration)
- SCGH MMG Sodium Nitroprusside
- SCGH MMG Terlipressin Acetate
- SCGOPHCG NPG Intravenous Therapy

16. High dose corticosteroid

High-dose corticosteroids are widely prescribed for their immunosuppressive and anti-inflammatory effects. The risk of adverse effects increases with both dosage and treatment duration. Prolonged high-dose use can lead to complications and suppress the hypothalamic-pituitary-adrenal (HPA) axis, requiring close monitoring.

High doses are generally defined as prednisolone above 50 mg/day or an equivalent.

Short, high-dose courses tend to cause fewer side effects than long-term lower doses and are usually well tolerated. Therapy lasting two weeks or less generally doesn't require tapering, as HPA suppression is unlikely within this timeframe.

When high-dose corticosteroids are necessary, they should be used for the shortest duration in line with current evidence. Tapering must carefully balance preventing disease flare-ups with avoiding cortisol deficiency symptoms from HPA suppression. Abrupt or rapid withdrawal can cause adrenal insufficiency symptoms.

The tapering approach depends on the clinical condition, guided by treatment protocols, symptom monitoring, and tolerability. Evidence for specific tapering regimens is limited.

When prescribing a high-dose corticosteroid, the prescriber should ensure an appropriate monitoring and following up plan including:

- Senior medical confirmation (registrar or consultant) for high-dose prescriptions by junior doctors lasting over two weeks.
- Documentation of a clear dose reduction plan in the medical record accessible to all treating clinicians.
- Patient education, including verbal counselling, written tapering instructions with specific dose change dates, and consumer medicines information.
- Clear instructions on follow-up care, specifying who will review the dose (e.g., clinic doctor or GP) and any pending test results needed for dose adjustment.
- If another clinician will manage the taper, provide them with a projected taper timeline, target dose, and monitoring parameters.
- Monitor for adverse effects based on dose/duration, such as blood sugar changes, sleep disturbances, stomach issues, and bone health.

Upon discharge:

- Include all of the above details in the discharge summary.
- Schedule discharges during clinical pharmacist hours when possible, ensuring pharmacist support for patient education and provision of written materials.